Certified Degree

in association with National Cyber Security Centre

Lancaster University

2025-2026 ASSESSMENTS

**Postgraduate**

## Masters Degree in Cyber Security – Individual Penetration Testing Challenge

**Available Time** **[6 Hours]**

**Recommended Completion Time** **[4 Hours]**

**Assessment Weighting** **[50%]**

## SCC.442 Penetration Testing

***Academic Honesty and Integrity***

*Students at Lancaster University are part of an academic community that values trust, fairness and respect and actively encourages students to act with honesty and integrity. It is a University policy that students take responsibility for their work and comply with the university's standards and requirements- found in the Manual of Academic Regulations and Practice. By submitting their answers students will be confirming that the work submitted is completely their own. Academic misconduct regulations are in place for all forms of assessment and students may familiarise themselves with this via the university website:*
*https://www.lancaster.ac.uk/academic-standards-and-quality/regulations-policies-and-committees/manual-of-academic-regulations-and-procedures/*

***Plagiarism***

*Plagiarism involves the unacknowledged use of someone else's work and passing it off as if it were one's own. This covers every form of submitted work, from written essays, video vignettes, and coding exercises. However, deliberately plagiarism with the intent to deceive and gain academic benefit is unacceptable. This is a conscious, pre-meditated form of cheating and is regarded as a serious breach of the core values of the University. More information may be found via the plagiarism framework website. All coursework is to be submitted electronically and will be run through our plagiarism detection mechanisms. Please ensure you are familiar with the University's Plagiarism rules and if you are in any doubt please contact your module tutor.*
*https://www.lancaster.ac.uk/academic-standards-and-quality/regulations-policies-and-committees/principles-policies-and-guidelines/plagiarism-framework/*

**AMBER CATEGORY**
Under this category, you are permitted to use AI tools in an assistive role as specified by the module tutor and required by the assessment.



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***General Guidance:***

You are required to conduct a penetration test on 5 machines and submit a separate report containing the results of the test to Moodle on Friday MV1.

Your submitted work MUST be written in a format that can be submitted to Moodle (.docx, .pdf, etc.). Your submission must be anonymous and should not contain your name or student number. A template has been provided and you are **strongly** recommended to use it. The submitted work must follow traditional penetration testing report styles (doing so through Microsoft Word, Open/Libre Office or LaTeX is recommended) such as the ones found here:

https://pentestreports.com/templates/downloads/PWKv1-Report.docx

https://github.com/robingoth/pentest-report-template

You have 2 hours (Friday Week 10 December 10am-12pm) to conduct a penetration test on the first 3 machines and a further 2 hours (Friday Week 10 December 1pm-3pm) on the remaining 2 machines to obtain proof flags via techniques and methodologies used during a traditional penetration test. **Access to these machines will only be available during this timeslot** (ensure that the results from this are recorded outside of MyLab for use during the second half of the assessment). **You will only have access to machines 1, 2 and 3 in the first half of the assessment and you will only have access to machines 4 and 5 in the second half of the assessment**.

You will be assessed on the following items:

- Number of flags obtained, and the steps taken to obtain the flags (70%)
    - Challenge 1 [9 Marks]
    - Challenge 2 [10 Marks]
    - Challenge 3 [10 Marks]
    - Challenge 4 [12 Marks]
    - Challenge 5 [15 Marks]
- Report Quality and Content (30%)

Different challenges have a different weighting of marks that generally corresponds to the difficulty in compromising them.

**Report Structure**

Your report needs to contain the following sections, you are strongly recommended to use the template provided.

- Section 1: An executive summary of the machines investigated and the vulnerabilities found. This should be short and convey a summary of the vulnerabilities found and the actions that should be taken to address them.
- Section 2: A discussion of the legal and ethical considerations of performing this penetration test.
- Sections 3-7: Details on the penetration test of each virtual machine.
- Section 8: Academic references.

We expect that you will use appropriate academic citations of any material you reference in this report.

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For each section describe your test of one vulnerable virtual machine, the following information should be provided within your report:

- A screenshot containing the contents of each proof.txt file that was obtained or other source of the captured flag.
- A description of the vulnerabilities that were exploited to obtain each flag.
- A detailed description of the methodology used for obtaining each flag, including screenshots and a written explanation of the methodologies used.
- Document (e.g., via screenshots) the commands that you executed and their outputs.
- If you used applications, document (e.g., via screenshots) how you used them and what their outputs were (e.g., via screenshots).
- A description of recommended mitigations for the identified vulnerabilities.

Your description of the actions you take to compromise the machine should be described step-by-step. The reader should be able to reproduce these attacks by following your report.

**GENERAL INSTRUCTIONS**

**Logging into Kali:**
Use username **kali** and password **kali**.

**Flags:**
Throughout this exercise there are marks available for different activities and discovering information. For all machines, completion of the task is denoted through the discovery of a flag. These flags are either contained within proof.txt files or provided directly and are of the form *442{Th1s_1s_4_Dummy_Fl4g!}*. If a **proof.txt** file exists, it will always be located in either the **user's home directory**, the **root directory (/root)**, or the **C:** drive of the machine. To gain the mark for a given flag you are required to submit a screenshot showing the contents of the flag alongside the methodology used to access it within your report.

Machines with a flag **THAT IS** contained within a proof.txt file: Challenge 1, Challenge 3, and Challenge 5.
Machines with a flag **THAT IS NOT** contained within a proof.txt file: Challenge 2 and Challenge 4.

**Progression:**
There is a total of 5 machines each containing 1 flag. There are no dependencies between the machines, and therefore you may attempt to test these machines in any order.

The following IP address is NOT part of the assessment and should **NOT** be attacked:

- 192.168.25.100

**Strategies:**
It is important to consider your strategy in performing a penetration test on these machines. You have limited time and some explorations that may be time consuming.

- If you need to perform a time-consuming search on one machine, move on and attempt to penetrate another machine. No scans should take more than a few minutes, if a scan has been running longer you should consider alternative parameters to give to the tool.

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- You may see multiple ways in which to attack the system. Focus on the simplest vector first, then try others if that fails.
- More marks are allocated for capturing a flag than basic tasks such as reconnaissance.

**Recommended Strategies:**
You are strongly recommended to first perform initial reconnaissance on all machines first. You should then focus on machines with a lower mark weight, as these will be (in general) easier than higher mark weight machines.

**Machine IP Addresses:**
All 5 of the machines are located within the same local network (192.168.25.0/24). You will need to identify their IP addresses individually to access them. You have been given permission to only perform a penetration test against this network.

**Generative AI Guidance**

In this assessment you are permitted to use AI tools in an assistive role. For example, this may be to:

- provide feedback on the text of your report for improvement,
- suggest potential ways in which the machine may be vulnerable, or
- suggest potential ways to exploit the machine.

This list is **not** exhaustive. If you are in doubt about if the ways in which you are using generative AI is or is not suitable, please get in touch with Matthew.

If you make use of Generative AI, then you must acknowledge the use of the Generative AI tool. To do this you should include a section in your report which contains:

- The name and version of the generative AI system used e.g., ChatGPT-4.0.
- The publisher (company that made the AI system) e.g., OpenAI.
- The URL of the AI system.
- A brief description (single sentence) for each of the contexts in which you used the tool.

Please be aware that use of Generative AI is **not** a substitute for appropriate academic and industrial references.

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## Learning Outcome Mapping (As Required by the External Certification Bodies)

| **Subject Specific Learning Outcomes: Knowledge, Understanding and Skills** | | |
|---|---|---|
| *Outcome* | *Covered* | *How is it covered* |
| Understand the penetration process, from utilising the human and physical element of a system, all the way through information gathering to an actual system penetration. | Yes | Students need to perform reconnaissance on systems and then attack those systems using specific vulnerabilities in order to obtain flags. |
| Apply strategies in order to compromise modern operating systems. | Yes | Students need to creatively devise approaches to compromise a system. |
| Describe a range of approaches to compromise modern network infrastructure. | Yes | Students will use multiple approaches (nmap, ssh, ftp) to interact with and compromise systems. |
| Be able to perform a series of zero-touch reconnaissance and information gathering process in order to facilitate and effective penetration. | Yes | Students will need to perform reconnaissance and information gathering on the vulnerable machines in order to vulnerabilities that can be used to compromise them. |
| Be apprised of cryptographic techniques including symmetric and asymmetric algorithms and block/stream ciphers. | Yes | Students need to be aware of if data is protected and how to crack that protection. |
| Understand common web infrastructure attack vectors and be able to utilise tools and skills in order to attack web infrastructures. | Yes | Students will exploit systems that use common web services such as SQL and FTP servers |
| Be apprised on common database vulnerabilities and attack approaches in order to create an effective attack against such systems. | Yes | Students will use SQL injection attacks to obtain flags |
| Have a detailed knowledge of approaches to evade current attack detection mechanisms deployed in networks. | Yes | Students need to consider how they interact with systems. |
| Be aware of the legal and ethical frameworks in which they must operate. | Yes | Students will need to write a short section discussing the legal and ethical aspects of performing a penetration test |
| **General Learning Outcomes: Knowledge, Understanding and Skills** | | |
| *Outcome* | *Covered* | *How is it covered* |
| Reason critically | Yes | Students will need to understand the results from reconnaissance and identify the suitable next steps. This involves critical thinking about the exploit paths to reach specific goals (in this case flags). |
| Apply relevant theoretical concepts | Yes | Students will need to apply their knowledge of how to perform the penetration test. |
| Identify and solve problems, both individually and working in groups and formulate appropriate methods for troubleshooting. | Partly | This is individual work and not group work. Students may undertake incorrect actions. They will need to identify where they went wrong and then identify the correct steps to take. |
| Demonstrate and exercise independence of mind and thought. | Yes | This work is individual, so students need to make their own decisions on how to perform the penetration test, including the order (or any parallelisation) of tasks. |
| Evaluate research and different types of information & evidence arguments critically. | | |
| Synthesise and select appropriate information from a number of sources. | | |
| Structure and communicate ideas effectively in writing | Yes | Students need to write a report, in which they need to clearly document and explain steps in which someone could reproduce the identified vulnerability. |
| Plan, undertake and report on an individual piece of research-based work. | Yes | Students will need to decide how to allocate a short period of time to test a large array of vulnerable machines, document evidence as they work through the machines and then write up reproducible steps to compromise the machines. |

## Assessment question summary table (As Required by the External Certification Bodies)

| **Question Number** | **Question type**<br>**K**: Knowledge | **Assigned marks /out of total marks per Q** |
|---|---|---|

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| | **C:** Comprehension<br>**A**: Application | |
|---|---|---|
| Q1-Q5 | KA | 70% |
| Q1 | KA | 9 Marks |
| Q2 | KA | 10 Marks |
| Q3 | KA | 10 Marks |
| Q4 | KA | 12 Marks |
| Q5 | KA | 15 Marks |
| Writeup | CA | 30% |

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# Report Marking Guidelines

| Topic (Suggested weight) | Distinction<br>70-100% | Very Good Pass<br>65-69% | Good Pass<br>60-65% | Pass<br>50-59% | Marginal Fail<br>45-49% | Clear Fail<br>0-44% |
|---|---|---|---|---|---|---|
| Critical understanding of topic (45) | Excellent understanding and exposition of relevant issues; insightful and well informed, clear evidence of independent thought; good awareness of nuances and complexities; appropriate use of theory | Clear awareness and exposition of relevant issues; some awareness of nuances and complexities but tendency to simplify matters; based on appropriate choice and use of theory | Shows awareness of issues and theories; attempts at analysis but tendency to lapse into description | Work shows understanding of topic but at superficial level; no more than expected from attendance at lectures; some irrelevant material; too descriptive | Establishes a few relevant points but superficial and confused; much irrelevant material; very little or no understanding of the issues raised by the topic or topic misunderstood; content largely irrelevant; no choice or use of theory; essay almost wholly descriptive; no grasp of analysis with many errors and/or omissions | |
| Structure of Research (15) | Substantial evidence of well implemented independent research<br>and / or<br>Substantial evidence of well selected evidence to support argument | Some evidence of independent research reasonably well implemented<br>and / or<br>Some evidence of identification of suitable evidence to support argument | | Insufficient evidence of independent research<br>and / or<br>very limited evidence used to support argument | No evidence of independent research<br>and / or<br>No attempt to identify suitable evidence to support argument | |
| Use of literature (5) | Excellent use of literature to support argument /points | Good use of literature to support arguments | Use of standard literature to support arguments | Use of secondary literature to support arguments | Relies on a superficial repeat of class notes | No significant reference to literature |
| Executive Summary (20) | Excellent; clear summary of what was found and how to address the issues | Very good; draws together main issue; some points identified to address | Reasonable summary; a few implications that need to be addressed | Executive summary does not do justice to body of report; too short; no direction on addressing issues | No recognisable executive summary | |
| Language (5) | Excellent; a delight to read | Carefully written; negligible errors | A few errors; generally satisfactory | Some errors; grammar and syntax needs attention | Frequent errors; needs urgent attention | |
| Structure and presentation (10) | Arguments clearly structured and logically developed; sensible weighting of parts; meaningful diagrams; properly formatted references | Arguments clearly structured and logically developed; good weighting of parts; meaningful diagrams; properly formatted references | Arguments reasonably clear but undeveloped; some meaningless diagrams or poor structure | Arguments not very clear; poor organisation of material; poor use of diagrams; poor referencing | Arguments often confused and undeveloped; no logical structure; very poor organisation of material; many meaningless diagrams; negligible referencing | |

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